

Track 8: The Feeder as a Power Plant

Progress report, in-progress concept, evolving through the event window

Team Idea_buzz | Jenil Charadva, Dhruv Jodhani, Jay Patel

Rooftop solar is becoming dense enough on some Gujarat feeders that midday generation will reverse power flow back into the grid, while evenings will still bring the usual demand peak. Thousands of homes will already have batteries and hybrid inverters that could absorb midday surplus and discharge during the evening peak, but each one will be tuned only for its own household, invisible to the DISCOM and uncoordinated with its neighbours. This submission will explore a concept level Virtual Power Plant for an 11 kV feeder of 200 to 500 such homes, a way to coordinate existing batteries to flatten the feeder's load curve without touching each household's backup reserve, along with a simple settlement mechanism for homeowners and a rough value case for the DISCOM.

1. The Problem, As We Understand It So Far

This is still an evolving read of the problem, but our current thinking is that this will not be a hardware gap, it will be a coordination gap. The batteries and inverters will already exist on these feeders, nobody will be orchestrating them yet. That will reframe the challenge from "deploy more storage" to "build a lightweight layer that makes existing storage visible and dispatchable", which will change what kind of solution is even worth exploring.

2. What Exists Today (and Where We Looked)

We looked at how this problem is being approached elsewhere before assuming ours was novel.

- **Tata Power and AutoGrid, Mumbai.** India's most mature live example, targeting 55,000 residential and 6,000 commercial and industrial customers for 75 MW of peak reduction, scaling to 200 MW. It will be built on behavioural and automated demand response, meaning it curtails load rather than dispatching stored energy.
- **DEWA VPP pilot, Dubai.** Closer to what we will build. It aggregates rooftop PV, electric vehicles, and two battery technologies using AI forecasting and orchestration software, delivering close to 3.3 MW of flexibility.
- **Next Kraftwerke, Germany.** The sector benchmark at scale, aggregating 5,000 units into a single dispatchable entity above 4,000 MW.
- **CSTEP's sector diagnosis for India.** Points out that the core hurdles will be the absence of a clear VPP regulatory framework and the high fixed cost of communication infrastructure for integrating storage, rather than the technology itself.
- **GERC net metering regulations for Gujarat.** Show that the state regulator already ties rooftop solar capacity to the connected distribution transformer's capacity, confirming that transformer level stress from solar is a recognised regulatory concern, not a hypothetical one.

- **Peer reviewed study on reverse power flow (Energies, 2022).** A simulation study of a real low voltage network found that beyond a certain PV penetration threshold, reverse power flow will push distribution transformers into an operating range that shortens their working life, giving a technical basis for why feeder level coordination will matter.

None of these will be built for our specific case, a single 11 kV feeder, consumer owned and mixed brand hardware, at a 200 to 500 home scale. That will be the space we plan to work in.

3. Where We Think the Real Gap Is

Reasoning from the above, our current view, and one we expect to revise, is that the gap will sit in five places. There will be no feeder level visibility into individual battery state, no common protocol across inverter brands, no settlement structure that pays homeowners without threatening their backup power, no trust layer that makes homeowners comfortable ceding any control, and no DISCOM facing quantification connecting battery coordination to money saved. We will treat each of these as a design constraint to be solved, not a problem to be solved away.

4. Our Current Concept

The shape of the idea right now is a closed loop coordination architecture, shown in the flow chart below.

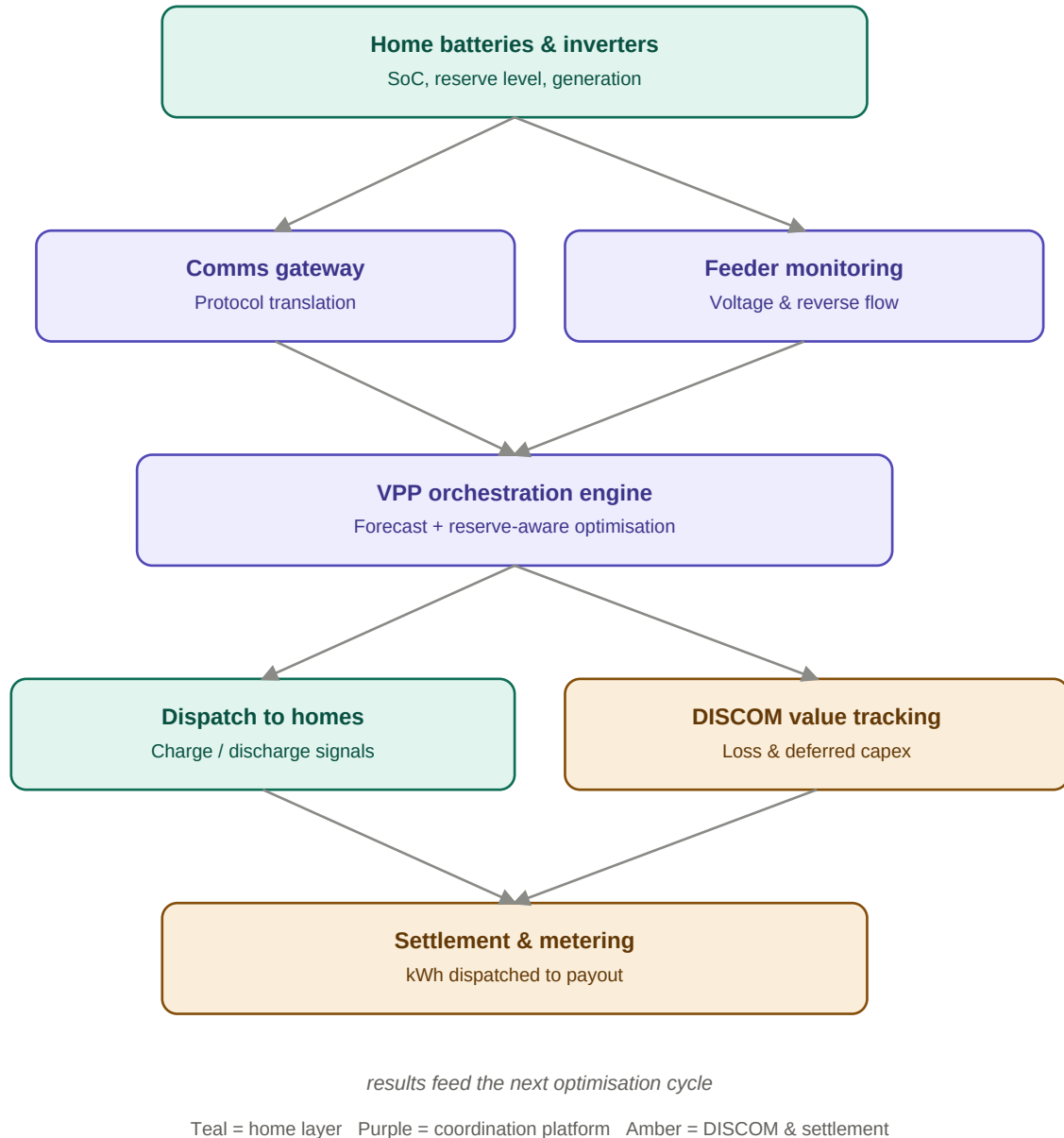


Figure 1. VPP coordination flow for the 11 kV feeder concept.

Home batteries and hybrid inverters will report their state (charge level, reserve setting, generation) through a communication gateway that will handle protocol translation across inverter brands. This will be combined with feeder level monitoring (voltage, reverse flow readings) inside a central VPP orchestration engine, which will run forecast driven, reserve aware dispatch optimisation. Its output will split two ways, dispatch commands back to homes that respect each household's own backup floor, and value data to the DISCOM covering loss and deferred capex tracking. Both paths will close through a settlement engine that pays homeowners for dispatched energy and feeds results into the next optimisation cycle.

This architecture is deliberately not final. The boxes will represent functions we believe need to exist, not a locked technical specification.

5. What We Haven't Cracked Yet

In the interest of representing this honestly as work in progress, three pieces will remain open problems we are actively working through, not gaps we are hiding.

- **Reserve aware dispatch logic.** The precise rule for how much of a home's stored energy will ever be available to the feeder, above its own backup floor.
- **Settlement formula.** How a homeowner's payout will be calculated from dispatched kWh, in a way that will be simple enough to explain and fair enough to trust.
- **DISCOM value case.** Turning "coordinated batteries reduce feeder stress" into a defensible rupee figure for deferred transformer or feeder upgrades and reduced technical losses at this scale.

References

1. Tata Power, "Tata Power announces Demand Response Program with AutoGrid," 2023. tatapower.com
2. Dubai Electricity and Water Authority, "DEWA completes its pilot Virtual Power Plant project," 2024. mediaoffice.ae
3. CSTEP, "The Future of Virtual Power Plants in India, A Perspective," 2023. cstep.in
4. Counterview, "Emerging technology of virtual power," 2023. counterview.in
5. Gujarat Electricity Regulatory Commission, "Net Metering Rooftop Solar PV Grid Interactive Systems (Fourth Amendment) Regulations," 2024. gercin.org
6. "Impact of Reverse Power Flow on Distributed Transformers in a Solar Photovoltaic Integrated Low Voltage Network," *Energies*, Volume 15, Issue 23, 2022.
7. PowerMonitors, "Reverse power flow in distribution feeders," Ask a Pro session transcript.